

FEATURES FORMED BY WATER

Some of the most obvious and striking features of Earth's surface are large bodies and flows of liquid water, such as oceans, seas, lakes, and rivers. In addition to these, there are landforms caused by the erosional or depositional power of liquid water, which include gorges, river valleys, and coastal features ranging from beaches to eroded headlands. Ice, too, has had a major impact on Earth's appearance. Ice-formed features include existing

GRAND CANYON

Carved over millions of years by the Colorado River, the Grand Canyon is Earth's largest gorge.

bodies of ice, such as glaciers and ice sheets, and landforms, such as U-shaped valleys, sculpted by the movement of past glaciers.

NORTH AMERICA *northeast*

Great Lakes

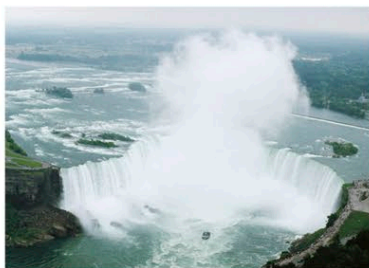


LOCATION Straddling the border of the US and Canada

TYPE System of freshwater lakes

AREA 94,480 sq. miles (244,767 sq. km)

The Great Lakes of North America are a system of five connected lakes that together form the largest body of fresh water on Earth. The lakes—named, from west to east, Superior, Michigan, Huron, Erie, and Ontario—contain 20 percent of Earth's surface fresh water and drain a basin of approximately 289,900 square miles (751,100 square km). They are connected to each other by short rivers, a strait, and canals and drain into the Atlantic Ocean via the St. Lawrence River. The Great Lakes began to form at the end of the last ice age when glacier-carved



NIAGARA FALLS

The greatest drop in water level within the Great Lakes system is at Niagara Falls, between lakes Erie and Ontario. Here, the water plunges a spectacular 167 ft (51 m).

basins were filled with meltwater left by a retreating ice sheet. Originally, several of today's lakes were united in one huge lake, but following post-glacial uplift in the region, they took on their present form about 10,000 years ago. The lake surfaces vary in height above sea level, from 600 ft (183 m) at Lake Superior to 246 ft (75 m) at Lake Ontario. Sprinkled across the lakes are thousands of islands, including Isle Royale on Lake Superior, which is itself big enough to hold several lakes.

SOUTH AMERICA *north*

Amazon River



LOCATION Flows from the Peruvian Andes, across Brazil to the Atlantic Ocean

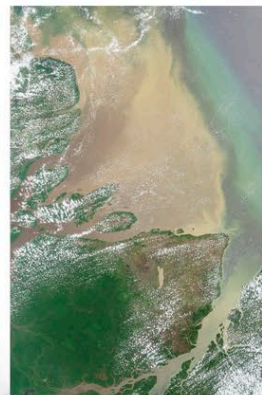
TYPE River

LENGTH 3,995 miles (6,430 km)

The Amazon is the greatest river on Earth, whether measured by the area of the planet's land surface that it drains or by the volume of water that it discharges every year. Overall, the Amazon accounts for nearly 20 percent of all river water discharged into Earth's oceans. The source of the Amazon has been established as a headwater of the Apurímac River, a tributary of the

MEETING THE ATLANTIC

The mouth of the Amazon occupies the whole top part of this image, which covers an area of tens of thousands of square miles. Rio Pará, the estuary of a separate major river, the Tocantins, can be seen at bottom.



BRAIDING

Over its course, the Amazon frequently braids into channels, creating many temporary islands.

Ucayali, high in the Andes of southern Peru. The Ucayali flows north from this area, turns east, and joins another major tributary, the Marañón River, where it becomes the Amazon proper. The river then meanders for thousands of miles across the Amazon Basin, a vast flat area that contains Earth's largest rainforest,

merging with numerous tributaries along the way. Just east of Manaus, at its confluence with the Negro River, the Amazon is already 10 miles (16 km) wide while still 1,000 miles (1,600 km) from the sea. At its mouth, the Amazon discharges into the Atlantic Ocean at the incredible rate of about 200 billion gallons of water (770 billion liters) every hour.



LAKES HURON AND SUPERIOR

In this photograph taken from a NASA Space Shuttle, the largest lake, Superior, is on the right, and appears partly iced over. Lake Huron is on the left.



MEANDERING TRIBUTARY

The Tigre is a tributary of the Amazon in Peru. Here, it meanders through the Peruvian rainforest, over 1,860 miles (3,000 km) from the Amazon's mouth.